

Problems marked with a (*) are a bit more complex.

12.1. Concatenation. Let $U \subset \mathbb{R}^n$ be open and connected and let $\gamma, \sigma \in C^1([0, 1], U)$ be paths such that $\gamma(0) = \sigma(1)$. Define the **concatenation of σ and γ** as the path

$$(\gamma * \sigma)(t) := \begin{cases} \sigma(2t) & \text{for } 0 \leq t \leq \frac{1}{2}, \\ \gamma(2t - 1) & \text{for } \frac{1}{2} \leq t \leq 1. \end{cases}$$

Prove rigorously that for all $F \in C^1(U, \mathbb{R}^n)$ it holds

$$\int_{\gamma * \sigma} F \cdot d(\gamma * \sigma) = \int_{\gamma} F \cdot d\gamma + \int_{\sigma} F \cdot d\sigma.$$

Extra: Explain what this formula means in terms of physical work.

12.2. The equation $\nabla f = F$. Let $U \subset \mathbb{R}^n$ be a simply connected domain and $F \in C^1(U, \mathbb{R}^n)$ satisfying the compatibility conditions $\frac{\partial F_i}{\partial x_j} = \frac{\partial F_j}{\partial x_i}$ for all $1 \leq i, j \leq n$. Fix some base point $p \in U$ and define $f : U \rightarrow \mathbb{R}$ as

$$f(x) := \int_{\gamma_x} F \cdot d\gamma_x, \text{ where } \gamma_x \in C^1([0, 1], U) \text{ is any path joining } p \text{ with } x.$$

1. Show that the definition of f is well-posed.
2. Show that if $\sigma \in C^1([0, 1], U)$ is a path joining x with x' then $f(x) - f(x') = \int_{\sigma} F \cdot d\sigma$.
3. Show that f is a potential for F , i.e. $F_i = \frac{\partial f}{\partial x_i}$ for all $1 \leq i \leq n$.
4. (*) With the same reasoning show that, even if we do not assume that U is simply connected, F admits a potential f in U if and only if for all closed loops $\gamma \in C^1(S^1, U)$ one has $\int_{\gamma} F \cdot d\gamma = 0$.

12.3. Work along paths and Stokes. Let $F : \mathbb{R}^3 \setminus (\{(0, 0)\} \times \mathbb{R}) \rightarrow \mathbb{R}^3$ be the vector field defined by

$$F(x, y, z) = \left(\frac{2(xz + y)}{x^2 + y^2}, \frac{2(yz - x)}{x^2 + y^2}, \log(x^2 + y^2) \right)$$

and let its $\gamma_1, \gamma_2, \gamma_3 : [0, 2\pi] \rightarrow \mathbb{R}^3$ be the paths

$$\begin{aligned} \gamma_1(t) &= (\cos(t), \sin(t), 0), \\ \gamma_2(t) &= (2 \cos(t), 2 \sin(t), 2), \\ \gamma_3(t) &= (3 \cos(t) + 6, 3 \sin(t), 3). \end{aligned}$$

1. Describe the domain of definition of F . Is it connected? Is it simply connected?
2. Compute the rotation $\text{rot}(F)$ of F and the path integral $\int_{\gamma_1} F \cdot d\gamma_1$ of F along γ_1 . Is F conservative?

3. Explain how the equation $\int_{\gamma_1} F \cdot d\gamma_1 = \int_{\gamma_2} F \cdot d\gamma_2$ follows from Stokes' theorem.
4. Does Stokes' theorem imply that $\int_{\gamma_1} F \cdot d\gamma_1 = \int_{\gamma_3} F \cdot d\gamma_3$ holds?

12.4. True or False. Let $U := \mathbb{R}^3 \setminus \{0\}$ and $F \in C^1(U, \mathbb{R}^3)$. For $r > 0$ set

$$I_r := \int_{\partial B_r} \langle F, \mathbf{n} \rangle dS$$

where $\mathbf{n}$ denotes the outward normal. Which of the following statements are always true?

- (a) If F is divergence-free, then $I_r = 0$ for all $r > 0$.
- (b) If F is divergence-free, then the value of I_r does not depend on r .
- (c) If F is curl-free, then the value of I_r does not depend on r .
- (d) If $F = \text{curl}(G)$ for a vector field $G \in C^1(U, \mathbb{R}^3)$ and $|F(x)| \leq C/|x|$, then $I_r = 0$ for all $r > 0$.

12.5. Conservative vector fields. For which values of $\lambda \in \mathbb{R}$ is the vector field $F: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by

$$F(x, y) = \begin{bmatrix} \lambda x e^y \\ (y + 1 + x^2)e^y \end{bmatrix}$$

conservative? Determine a potential for v for these values.

12.6. Reality check on Stokes. The upper hemisphere $S = \{(x, y, z) \in \mathbb{S}^2 \mid z \geq 0\}$ of $\mathbb{S}^2$ is oriented by the normal field $n: S \rightarrow \mathbb{R}^3$ with positive z -component, and let $F: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be the vector field given by

$$F(x, y, z) = (xy, z^2, 1).$$

Calculate the flux $\int_S \text{rot}(F) \cdot dn$ of $\text{rot}(F)$ through S in two ways:

1. directly using the definition of flux integrals,
2. using Stokes' theorem.

12.7. Some identities. Let $f: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be a vector field that is twice continuously differentiable, and let $\varphi: \mathbb{R}^3 \rightarrow \mathbb{R}$ be a C^2 function. We remind that we denote by $\nabla\varphi$ the gradient of φ . Prove the following identities.

1. $\text{rot}(\varphi f) = (\nabla\varphi) \times f + \varphi \text{rot} f$,
2. $\text{div}(\varphi f) = \langle \nabla\varphi, f \rangle + \varphi \text{div} f$,
3. $\text{div}(\text{rot} f) = 0$,
4. $\text{rot}(\nabla\varphi) = 0$.

12.8. Multiple Choice. Let $Q = [a, b] \times [c, d] \subset \mathbb{R}^2$ and $f \in C(Q, \mathbb{R}^2)$. Let $\mathbf{n}$ be an outward normal field on Q and

$$I_b = \int_{[a,b] \times \{c\}} \langle f, \mathbf{n} \rangle, \quad I_t = \int_{[a,b] \times \{d\}} \langle f, \mathbf{n} \rangle$$

$$I_l = \int_{\{a\} \times [c,d]} \langle f, \mathbf{n} \rangle, \quad I_r = \int_{\{b\} \times [c,d]} \langle f, \mathbf{n} \rangle$$

Which of the following expressions corresponds to the outer flux of f through ∂Q ?

(a) $I_b - I_t + I_r - I_l$

(b) $I_b + I_t + I_r + I_l$

(c) $I_b + I_t - I_r - I_l$

(d) $-I_b + I_t - I_r + I_l$

12.9. Change of variables with differential forms. Let $U, V \subset \mathbb{R}^n$ be open sets, and let $\psi = (\psi^1, \dots, \psi^n) : U \rightarrow V$ be a diffeomorphism with $\det(\frac{\partial \psi_j}{\partial x_i}) > 0$. Define an n -differential form ω on U by $\omega = d\psi^1 \wedge d\psi^2 \wedge \dots \wedge d\psi^n$.

(a) Show that $\omega = \det(\frac{\partial \psi_j}{\partial x_i}) dx^1 \wedge \dots \wedge dx^n$

(b) From the relation in (a), rewrite the change of variables formula using differential forms.